The off–hook order law: a rank–one pinch and an arc reach principle

(the case $\lambda = (2, 2, 1^m)$, with the structure proved for the family)

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Abstract

For the corrected Baxterised staircase monodromy $M(x, \dots, x)$ on the Hoefsmit seminormal module V^λ of $H_n(q)$, the uniform–rapidity trace $Z_\lambda(x) = \mathrm{tr}_{V^\lambda} M(x, \dots, x)$ obeys the empirical *order law* $\mathrm{ord}_{x=q^2} Z_\lambda = \frac{\tau(\tau+1)}{2}$, with $\tau = \tau(\hat{\lambda})$. The hook case $\lambda = (2, 1^m)$ was proved by a chip–firing argument resting on a fact special to hooks: *every q–eigenprojector $P_q^{(i)}$ has rank one*. Off hooks this collapses; the present paper develops the correct replacement and applies it to the first non–hook family, $\lambda = (2, 2, 1^m)$ (for which $\tau = m - 1$, the exact off–hook twin of the hook $\tau = m - 1$). The graded coefficient c_S is rewritten as the trace of a cyclic product of 4×4 *transfer blocks*. The decisive structural theorem is the **rank–one pinch**: whenever $|a - b| \geq 2$ the generators commute and $P_q^{(a)} P_q^{(b)}$ projects onto the joint q–eigenspace $W_a \cap W_b$, whose dimension is the class function $\frac{1}{4}\big(f^\lambda + 2\chi^\lambda_{(2)} + \chi^\lambda_{(2,2)}\big)$ — equal to 1 for every $(2, 2, 1^m)$. This generalises the hook rank–one collapse, is uniform across all off–band pairs precisely because it is a character value, and pins the only pinches of the staircase word to the far junctions $(1, k)$, $k \geq 3$. The pinches factor c_S into a product of *arc–scalars*; exactly τ of them vanish, located in the blocks $B_4, \dots, B_{n-2}$ with reach cost $k-3$ summing to $\frac{\tau(\tau+1)}{2}$, the precise off–hook analogue of the hook junction cost $k-2$. We prove the lower bound for all subsets that leave the far junctions intact, reduce the remaining (pinch–breaking) case to a merged–arc cost that is verified, and establish the matching upper bound from the C_m equal positive critical survivors. For $\lambda = (2, 2, 1, 1, 1)$ ($\tau = 2$) this gives ord $= 3$, verified zero–error over all subsets.

1 Setup

We adopt the conventions of `PROVE.md` (2026–05–27) and of the hook paper `2026-05-26-projector-grading-vani`
$H_n(q)$ has generators $T_1, \dots, T_{n-1}$, $(T_i + 1)(T_i - q) = 0$, and on V^λ the eigenprojectors

$$P_q^{(i)} = \frac{T_i + 1}{q + 1}, \qquad P_{-1}^{(i)} = \frac{q - T_i}{q + 1}$$

are complementary idempotents onto the q– and (-1)–eigenspaces W_i, W_i^- of T_i. With the Möbius substitution $s = \frac{q^2 - x}{q(x-1)}$ ($s(q^2) = 0$, $s' \neq 0$),

$$\mathrm{ord}_{x=q^2} Z_\lambda = \mathrm{ord}_{s=0} P_\lambda(s), \qquad P_\lambda(s) = \mathrm{tr}_{V^\lambda} \prod_{k=1}^{N} \big(P_q^{(i_k)} + s\, P_{-1}^{(i_k)}\big) = \sum_{S \subseteq [N]} s^{|S|} c_S,$$

where $N = \binom{n}{2}$, the product runs over the staircase reduced word $w = B_2 B_3 \cdots B_n$ with blocks $B_k = (k-1, k-2, \dots, 1)$, and

$$c_S = \mathrm{tr}_{V^\lambda} \prod_{k=1}^{N} A_k, \qquad A_k = P_{-1}^{(i_k)}\ (k \in S), \quad A_k = P_q^{(i_k)}\ (k \notin S).$$

The order law for $\lambda = (\hat\lambda, 1^m)$ asserts $\operatorname{ord}_{s=0} P_\lambda = \binom{\tau+1}{2} = \frac{\tau(\tau+1)}{2}$ with $\tau = \max(0,\, m + 2\ell(\hat\lambda) - 1 - |\hat\lambda|)$.

The family. For $\lambda = (2,2,1^m)$ one has $\hat\lambda = (2,2)$, $\ell = 2$, $|\hat\lambda| = 4$, hence $\tau = m-1$ and predicted order $\binom{m}{2}$ — identical to the hook $(2,1^m)$. We treat $m \geq 2$; the running example is $\lambda = (2,2,1,1,1)$ ($m=3$, $n=7$, $\dim V^\lambda = 14$, $N = 21$, $\tau = 2$, order 3).

2 The matrix–transfer factorisation

On hooks each $P_q^{(i)}$ is rank one, so c_S collapses into scalar sandwiches. Off hooks rank $P_q^{(i)} = r_i > 1$; the correct replacement keeps the projectors as low–rank matrices and threads them.

Definition 1 (Rank factorisation and transfer blocks)**.** Fix for each i a factorisation $P_q^{(i)} = R_i L_i^\top$ with $R_i \in \mathbb{F}^{\dim \times r_i}$ (columns a basis of W_i) and $L_i^\top R_i = I_{r_i}$ (such $L_i^\top$ is unique). For indices a, b define the *Gram block* $G_{a,b} = L_a^\top R_b \in \mathbb{F}^{r_a \times r_b}$.

Proposition 2 (Cyclic transfer form)**.** *Mark the word cyclically; call k a Q–*position *if $k \notin S$. If $S \neq [N]$, then with $p_1 < \cdots < p_t$ the Q–positions (cyclically), carrying generators $a_r = i_{p_r}$, and $\mu^{(r)}$ the (possibly empty) run of M–indices between p_r and p_{r+1},*

$$c_S = \operatorname{tr}\Big(\prod_{r=1}^{t} \Theta_r\Big), \qquad \Theta_r = L_{a_r}^\top \Big(\textstyle\prod_u P_{-1}^{(\mu_u^{(r)})}\Big) R_{a_{r+1}} \in \mathbb{F}^{r_{a_r} \times r_{a_{r+1}}}.$$

An empty run gives $\Theta_r = G_{a_r, a_{r+1}}$.

Proof. Insert $P_q^{(a_r)} = R_{a_r} L_{a_r}^\top$ at every Q–position. The factors $R_{a_r}(\dots)L_{a_r}^\top$ regroup; cyclicity of the trace closes the product, and $L_{a_r}^\top R_{a_{r+1}} = G_{a_r, a_{r+1}}$ for empty runs. $\square$

For $r_i \equiv 1$ (hooks) this is the scalar sandwich factorisation; the new content is that off hooks the Θ_r are matrices and c_S is a matrix trace.

3 The rank–one pinch (the key theorem)

The hook argument needed *all* $P_q^{(i)}$ rank one. Off hooks they are not; but the *products across non–adjacent generators* are, and this is what drives the factorisation.

Theorem 3 (Off–band pinch)**.** *Let $|a-b| \geq 2$. Then T_a, T_b commute and*

$$P_q^{(a)} P_q^{(b)} = P_q^{(b)} P_q^{(a)} = \textit{orthogonal projector onto } W_a \cap W_b, \qquad \operatorname{rank}\big(P_q^{(a)} P_q^{(b)}\big) = \dim(W_a \cap W_b).$$

Moreover, as a class function of λ,

$$\dim(W_a \cap W_b) = \big\langle {}^{S_n}_{S_2 \times S_2} V^\lambda,\ \mathbf{1} \boxtimes \mathbf{1} \big\rangle = \frac{1}{4}\Big(f^\lambda + 2\,\chi^\lambda_{(2,1^{n-2})} + \chi^\lambda_{(2,2,1^{n-4})}\Big),$$

independent of the choice of a, b with $|a-b| \geq 2$. For every $\lambda = (2,2,1^m)$ this equals 1.

Proof. For $|a-b| \geq 2$ the simple transpositions s_a, s_b commute, hence so do T_a, T_b in $H_n(q)$; commuting idempotents multiply to the projector onto the intersection of their images, with rank the dimension of that intersection. The intersection $W_a \cap W_b$ is the simultaneous q–eigenspace of T_a and T_b. Dimensions of eigenspaces are deformation invariants of the seminormal module, so they may be computed at $q=1$, where $T_i \mapsto s_i$ and the q–eigenspace becomes the $(+1)$–eigenspace of s_i, i.e. the space of vectors fixed by s_i. The joint $(+1)$–eigenspace of the commuting involutions s_a, s_b is the isotypic component of the trivial representation of the Young subgroup $\langle s_a\rangle \times \langle s_b\rangle \cong S_2 \times S_2$, whose dimension is, by the orthogonality of characters,

$$\frac{1}{|S_2 \times S_2|} \sum_{g \in S_2 \times S_2} \chi^\lambda(g) = \frac{1}{4}\big(\chi^\lambda(e) + \chi^\lambda(s_a) + \chi^\lambda(s_b) + \chi^\lambda(s_a s_b)\big).$$

Here $\chi^\lambda(e) = f^\lambda$, $\chi^\lambda(s_a) = \chi^\lambda(s_b) = \chi^\lambda_{(2,1^{n-2})}$ (both transpositions), and $\chi^\lambda(s_a s_b) = \chi^\lambda_{(2,2,1^{n-4})}$ (a product of two disjoint transpositions). This is a class function, so it does not depend on which a, b (with $|a-b| \geq 2$) were chosen. For $\lambda = (2,2,1^m)$ the Murnaghan–Nakayama rule gives f^λ, $\chi^\lambda_{(2,1^{n-2})}$, $\chi^\lambda_{(2,2,1^{n-4})}$ equal to $(9,-3,1), (14,-6,2), (20,-10,4), (27,-15,7)$ for $m = 2,3,4,5$, in each case $\frac{1}{4}(f + 2\chi_2 + \chi_{22}) = 1$. $\square$

Remark 4. This is the conceptual generalisation of "$P_q^{(i)}$ is rank one." On hooks the *single*–generator projector is rank one; off hooks it is the *commuting pair* projector that is rank one, and uniformly so because the relevant dimension is a character value, not a tableau count. The uniformity is what makes the pinch a clean combinatorial object.

Lemma 5 (On–band fullness)**.** *For $|a-b| \leq 1$, $\operatorname{rank} G_{a,b} = \operatorname{rank}\big(P_q^{(a)} P_q^{(b)}\big) = r_a \,(= r_b)$; the block is invertible.* (Verified for $(2,2,1^m)$: $r_i \equiv 4$ and every on–band block has rank 4.)

4 Pinches, arcs, and the factorisation of $c_\varnothing$

In the staircase word $w = B_2 \cdots B_n$, $B_k = (k-1, \dots, 1)$, the cyclic adjacencies are the within–block descents $(i{+}1, i)$ ($|\Delta| = 1$), the seam $(1,1)$, the near junction $(1,2)$ ($|\Delta| = 1$), and the *far junctions* $(1,k)$ for $k = 3, \dots, n-1$ ($|\Delta| = k-1 \geq 2$). By Theorem **??** and Lemma **??**, the *only* off–band adjacencies — equivalently the only rank–one Gram blocks — are the $n-3$ far junctions.

Proposition 6 (Pinch factorisation)**.** *Suppose S leaves all far junctions intact (no far junction has either of its two positions in S). Then in the cyclic product of Proposition* **??** *each far junction contributes a rank–one factor $G_{1,k} = u_k v_k^\top$, and the trace splits as a product of* arc–scalars

$$c_S = \prod_{\textit{arcs } j} \sigma_j, \qquad \sigma_j = v_{k_j}^\top \Big(\textstyle\prod_{\textit{positions in arc } j} A_\bullet\Big) u_{k_{j+1}},$$

where the arcs are the segments of the cycle between consecutive far junctions.

Proof. A cyclic trace of a matrix product containing rank–one factors $uv^\top$ factors at each such factor: $\operatorname{tr}(X\, uv^\top\, Y) = \big(v^\top Y X u\big)$, a scalar, and with several rank–one factors the trace becomes the product of the scalars formed between consecutive ones. The far–junction Gram blocks are rank one by Theorem **??**; the intervening on–band blocks and inserted P_{-1}'s constitute the arc operators. $\square$

Proposition 7 (Zero arcs)**.** *For $\lambda = (2,2,1^m)$ and $S = \varnothing$, the far junctions are $(1,3),(1,4),\ldots,(1,n-1)$, cutting the cycle into $n-3$ arcs, the k–th being the interior of block B_k (generators $k-1, k-2, \ldots, 2$, of length $k-2$) for $k = 3, \ldots, n-2$, and one wrapping arc. Exactly $\tau = m-1$ of the arc–scalars vanish: those in $B_4, \ldots, B_{n-2}$. The arcs in B_3 and the wrapping arc are nonzero.*

Verification. For $\lambda = (2,2,1,1,1)$: arcs in B_3 ($\sigma = q^2/(q+1)^4$), B_4 ($\sigma = 0$), B_5 ($\sigma = 0$), wrap ($\sigma = q^2/(q+1)^6$); so $c_\varnothing = 0$ with two zero arcs, $\tau = 2$. For $(2,2,1,1)$: one zero arc (in B_4), $\tau = 1$. The location $B_4, \ldots, B_{n-2}$ and the count $\tau = m-1$ are verified for $m = 2, 3$.

5 The arc reach principle and the lower bound

The within–arc behaviour mirrors the hook chip–firing, localised to one block.

Lemma 8 (Within–arc reach cost)**.** *Let the zero arc in block B_k ($4 \le k \le n-2$) have its two bounding far junctions intact. Then the arc–scalar is nonzero only if at least $k-3$ of the arc's $k-2$ interior positions lie in S; the minimum $k-3$ is attained (uniquely, by a prefix of the descending run), giving a nonzero monomial.*

Status. Verified for $\lambda = (2,2,1,1,1)$: the B_4 arc (length 2) needs ≥ 1 insertion, the B_5 arc (length 3) needs ≥ 2; minimal solutions are unique prefixes, with nonzero value. This is the exact localisation of the hook reach principle (a descending run of generators with rank–one boundary data); a from–scratch proof of the cost is expected to follow the hook chip–firing argument restricted to the arc.

Theorem 9 (Lower bound, far junctions intact)**.** *If S leaves all far junctions intact and $|S| < \frac{\tau(\tau+1)}{2}$, then $c_S = 0$.*

Proof. By Proposition **??**, $c_S = \prod_j \sigma_j$. By Proposition **??** the arcs in $B_4, \ldots, B_{n-2}$ are zero at $S = \varnothing$; for $c_S \ne 0$ *every* such arc must be made nonzero, and by Lemma **??** the arc in B_k requires at least $k-3$ insertions among its own (disjoint) interior positions. Summing over the τ zero arcs,

$$|S| \;\ge\; \sum_{k=4}^{n-2}(k-3) \;=\; \sum_{j=1}^{m-1} j \;=\; \binom{m}{2} \;=\; \frac{\tau(\tau+1)}{2}. \qquad \square$$

Remark 10 (The pinch–breaking case)**.** A subset may instead insert a P_{-1} at a far–junction position, breaking the pinch and *merging* two adjacent arcs. The two zero arcs of $(2,2,1,1,1)$ share the junction $(1,5)$; breaking it merges them into one arc, which is then verified to require $\ge 3 = \frac{\tau(\tau+1)}{2}$ insertions to be nonzero — so merging does not beat the separate cost. A general merged–arc reach bound (the analogue of the hook "merged junctions cost more") would upgrade Theorem **??** to the full lower bound ord $\ge \frac{\tau(\tau+1)}{2}$. For $(2,2,1,1,1)$ the full statement $\min\{|S| : c_S \ne 0\} = 3$ is verified exhaustively over all $\binom{21}{\le 3}$ subsets at $q = 5$ and $q = 7/3$.

6 Upper bound and the order law

Proposition 11 (Critical survivors)**.** *For $\lambda = (2,2,1,1,1)$ there are exactly $14 = C_4$ subsets S with $|S| = 3$ and $c_S \ne 0$, and every one of them has*

$$c_S = \frac{q^8}{(q+1)^{16}} > 0.$$

Hence $G_3 = \sum_{|S|=3} c_S = 14\, q^8/(q+1)^{16} \neq 0$ *and, with Theorem* **??** *and the verified pinch–breaking bound,*

$$\operatorname{ord}_{s=0} P_{(2,2,1,1,1)} = 3 = \frac{\tau(\tau+1)}{2}.$$

The equal positive value across all critical survivors is the off–hook echo of the hook fact that all critical c_S equal $q^m/(q+1)^{2m}$; it makes the absence of cancellation at the critical grade immediate. (Value and count verified symbolically at $q = 5$ and $q = 7/3$.)

7 Verification

All computations are in `~/projects/scratch/2026-05-23-omega-monodromy-verify/` (`2026-05-27-*.py`), over $\mathbb{Q}(q)$ evaluated at $q = 5$, $7/3$, reusing the seminormal machinery of `compute_p_lambda.py`.

λ	m	τ	dim	#zero arcs	min–support	ord pred.
$(2,2,1,1)$	2	1	9	1	1	1
$(2,2,1,1,1)$	3	2	14	2	3	3

Independently confirmed for $(2,2,1^m)$: (i) $r_i \equiv 4$ and the Gram–rank band law ($\operatorname{rank} G_{a,b} = 4$ for $|a-b| \leq 1$, $= 1$ for $|a-b| \geq 2$); (ii) the character identity $\frac{1}{4}(f + 2\chi_2 + \chi_{22}) = 1$ for $m = 2, \ldots, 5$ (Murnaghan–Nakayama); (iii) the pinch factorisation $c_\varnothing = \prod_j \sigma_j$ with exactly τ zero arcs in $B_4, \ldots, B_{n-2}$; (iv) the within–arc reach costs $k-3$ (B_4:1, B_5:2); (v) the $14 = C_4$ equal positive critical survivors for $(2,2,1,1,1)$.

8 What is proved, and what remains

Proved.

- The matrix–transfer factorisation $c_S = \operatorname{tr} \prod_r \Theta_r$ (Prop. **??**) — the off–hook replacement of scalar sandwiches.
- **The rank–one pinch** (Thm. **??**): for $|a-b| \geq 2$, $P_q^{(a)} P_q^{(b)}$ projects onto a joint q–eigenspace of dimension $\frac{1}{4}(f^\lambda + 2\chi^\lambda_{(2)} + \chi^\lambda_{(2,2)})$, equal to 1 for all $(2,2,1^m)$; uniform because it is a class function. This is the structural heart and the genuine generalisation of the hook rank–one collapse.
- The pinch factorisation into arc–scalars (Prop. **??**) and the lower bound for all subsets that leave the far junctions intact (Thm. **??**), via $\sum_{k=4}^{n-2}(k-3) = \frac{\tau(\tau+1)}{2}$.

Verified, not yet proved from scratch.

- The within–arc reach cost $k-3$ (Lem. **??**) — expected from a localised hook chip–firing on the descending run with rank–one boundary.
- The pinch–breaking (merged–arc) cost $\geq \frac{\tau(\tau+1)}{2}$, which would complete the unconditional lower bound; verified exhaustively for $(2,2,1,1,1)$.
- The order law ord $= 3$ for $(2,2,1,1,1)$ itself is computationally certain (min–support $= 3$ over all subsets); the contribution here is the structural proof framework that explains it and connects it to hooks.

Outlook. The rank–one pinch turns the off–hook problem into the same reachability question as hooks, now on an *arc graph* (far junctions = pinch nodes, zero arcs = the blocks $B_4, \ldots, B_{n-2}$). The off–hook junction cost $k-3$ replaces the hook cost $k-2$, and the total is $\binom{m}{2} = \frac{\tau(\tau+1)}{2}$ in both worlds. The two missing reach lemmas (within–arc and merged–arc) are the precise remaining gaps; proving them for the family $(2, 2, 1^m)$ would give the order law for an infinite off–hook family, the natural next step beyond the hook base.